

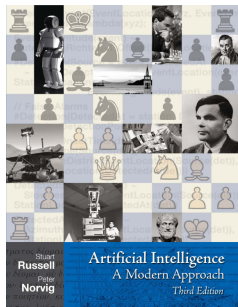
Machine Learning

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Chapter 1: Probabilities

- **Probability basics**
- **Some important probability distributions**
- **Bayes rule and conditional inference**
- **Origins of probabilities: Bayesian and frequentist interpretation**



Probabilities

Definition (Probability Space)

A probability space is the triple

$$(\Omega, S, P)$$

where

- Ω is the sample/outcome space, $\omega \in \Omega$ is a **sample point/atomic event**.

Example: 6 possible rolls of a die: $\Omega = \{1, 2, 3, 4, 5, 6\}$

- S is a collection of **events** to which we are willing to assign probabilities. An **event** $a \in S$ is any subset of Ω , e.g., die roll < 4 :
 $a = \{1, 2, 3\}$
- P is a mapping from events in S to $\mathbb{R}$ that satisfies the probability axioms.

Axioms of Probability

- ① $P(a) \geq 0 \forall a \in S$: probabilities are not negative,
- ② $P(\Omega) = 1$: “trivial” event has maximal possible prob 1,
- ③ $a, b \in S$ and $a \cap b = \{ \}$ $\Rightarrow P(a \cup b) = P(a) + P(b)$: probability of two mutually disjoint events is the sum of their probabilities.

Example:

$$P(\text{die roll} < 4) = P(1) + P(2) + P(3) = 1/6 + 1/6 + 1/6 = 1/2.$$

Random Variables

Definition (Random Variable)

A **random variable** X is a function from the sample points to some range, e.g., the reals

$$X : \Omega \rightarrow \mathbb{R},$$

or booleans

$$X : \Omega \rightarrow \{\text{true}, \text{false}\}.$$

Real random variables are characterized by their distribution function.

Definition (Cumulative Distribution Function)

Let $X : \Omega \rightarrow \mathbb{R}$ be a real valued random variable. We define

$$F_X(x) = P(X \leq x).$$

This is the probability of the event $\{\omega \in \Omega : X(\omega) \leq x\}$

Boolean RVs and propositional logic

- Dentistry example: **Boolean** random variable (dental) *Cavity*
- Proposition: answer to question “do I have a cavity?”
 $\rightsquigarrow$ *Cavity = true is a proposition, also written cavity*
- **Proposition**: event (=set of sample points / atomic events) where the **proposition is true**.
- Given Boolean random variables A and B :
 - ▶ event a = set of atomic events where $A(\omega) = \text{true}$
 - ▶ event $\neg a$ = set of atomic events where $A(\omega) = \text{false}$
 - ▶ event $a \wedge b$ = atomic events where $A(\omega) = \text{true}$ and $B(\omega) = \text{true}$
- With Boolean variables, **event = propositional logic model**
e.g., $A = \text{true}$, $B = \text{false}$, or $a \wedge \neg b$.

Proposition = disjunction of events in which it is true

e.g., $(a \vee b) = (\neg a \wedge b) \vee (a \wedge \neg b) \vee (a \wedge b)$

$\implies P(a \vee b) = P(\neg a \wedge b) + P(a \wedge \neg b) + P(a \wedge b)$

Dentistry example: Cavity and Toothache

	<i>Toothache = true</i>	<i>Toothache = false</i>
<i>Cavity = true</i>	0.12	0.08
<i>Cavity = false</i>	0.08	0.72

For any proposition ϕ , sum the atomic events where it is true:

$$P(\text{Cavity} = \text{true}) = P(\text{cavity}) =$$

$$P(\text{cavity} \wedge \text{toothache}) + P(\text{cavity} \wedge \neg \text{toothache}) = 0.12 + 0.08 = 0.2$$

$$P(\text{Toothache} = \text{false}) = P(\neg \text{toothache}) = 0.08 + 0.72 = 0.8$$

$$\begin{aligned} P(\text{cavity} \vee \text{toothache}) &= P(\neg \text{cavity} \wedge \text{toothache}) \\ &\quad + P(\text{cavity} \wedge \neg \text{toothache}) + P(\text{cavity} \wedge \text{toothache}) \\ &= 0.08 + 0.08 + 0.12 = 0.28. \end{aligned}$$

Syntax for Propositions

- **Boolean** random variables
e.g., *Cavity* (do I have a cavity?)
Cavity = *true* is a proposition, also written *cavity*
- **Discrete** random variables (finite or infinite)
e.g., *Weather* is one of $\langle \textit{sunny}, \textit{rain}, \textit{cloudy}, \textit{snow} \rangle$
Weather = *rain* is a proposition
Values must be exhaustive and mutually exclusive
- **Continuous** random variables (bounded or unbounded)
e.g., *Temp* = 21.6; also allow, e.g., *Temp* < 22.0.

Probability distribution

- **Unconditional probabilities** of propositions
e.g., $P(\text{Weather} = \text{sunny}) = 0.72$.

Bayesian interpretation:

Belief, prior to arrival of any (new) evidence

- **Probability distribution** gives values for all possible assignments:
 $P(\text{Weather}) = \langle 0.72, 0.1, 0.08, 0.1 \rangle$ (normalized, sums to 1)
- **Joint probability distribution** for a set of RVs gives the probability of every atomic event on those RVs:
 $P(\text{Weather}, \text{Cavity}) =$ a 4×2 matrix of values:

<i>Weather =</i>	<i>sunny</i>	<i>rain</i>	<i>cloudy</i>	<i>snow</i>
<i>Cavity = true</i>	0.144	0.02	0.016	0.02
<i>Cavity = false</i>	0.576	0.08	0.064	0.08

Probability for continuous variables

Suppose X describes some uncertain continuous quantity.

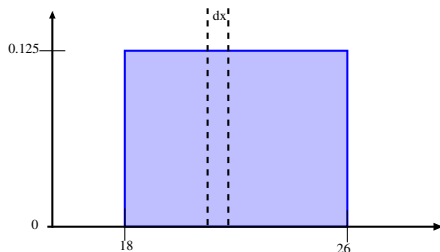
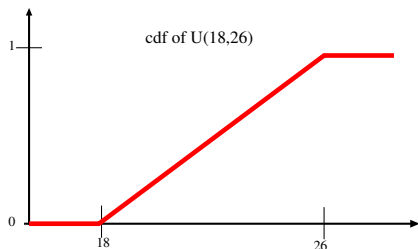
What is the probability of the event $a < X \leq b$?

- define events $A = (X \leq a)$, $B = (X \leq b)$, $W = (a < X \leq b)$.
- $B = A \vee W$. A and W are mutually exclusive
 $\leadsto P(B) = P(A) + P(W) \leadsto P(W) = P(B) - P(A)$.
- Define the **cumulative distribution function (cdf)** as
 $F(q) := P(X \leq q)$: $P(W) = P(B) - P(A) = F(b) - F(a)$.
- Assume that F is absolutely continuous:
define **probability density function (pdf)** $p(x) := \frac{d}{dx} F(x)$.
- Given a **pdf**, the probability of a continuous variable being in a **finite interval** is: $P(a < X \leq b) = \int_a^b p(x) dx$.
- As the size of the interval gets smaller, we can write
 $P(x < X \leq x + dx) \approx p(x) dx \Rightarrow p(x) \approx P(x < X \leq x + dx)/dx$.
Finally: $p(x) = \lim_{dx \rightarrow 0} P(x < X \leq x + dx)/dx$.
- Note: $p(x) \geq 0$ and $\int_{-\infty}^{+\infty} p(x) dx = 1$, but $p(x) > 1$ is possible.

Probability for continuous variables

Example: uniform distribution:

$$\text{Unif}(a, b) = \frac{1}{b - a} \mathbb{I}(a \leq x \leq b).$$



$p(X = 20.5) = 0.125$ really means

$$\lim_{dx \rightarrow 0} P(20.5 < X \leq 20.5 + dx) / dx = 0.125$$

Mean and Variance

- Most familiar property of a distribution: **mean**, or **expected value**, denoted by μ or $E[X]$.

- Discrete RVs:

$$E[X] = \sum_{x \in \mathcal{X}} xp(x),$$

- Continuous RVs:

$$E[X] = \int_{\mathcal{X}} xp(x) dx.$$

If this integral is not finite, the mean is not defined.

- The **variance** is a measure of the **spread** of a distribution:

$$\begin{aligned} \text{var}[X] &:= \sigma^2 = E[(X - \mu)^2] = E[X^2 - 2X\mu + \mu^2] \\ &= E[X^2] - 2\mu E[X] + \mu^2 = E[X^2] - 2\mu^2 + \mu^2 = E[X^2] - \mu^2 \end{aligned}$$

- The square root $\sqrt{\text{var}[X]}$ is the **standard deviation**.

Conditional probability

- **Conditional** probabilities, e.g., $P(\text{cavity}|\text{toothache}) = 0.8$
“Prob. of cavity is 80%, **given that toothache is all I know**”
NOT “if *toothache* then 80% chance of *cavity*”
- Notation for conditional distributions:
 $P(\text{Cavity}|\text{Toothache}) = 2\text{-element vector of 2-elem. vectors.}$
- If we **know more**, e.g., *cavity* is also given, then we have
 $P(\text{cavity}|\text{toothache}, \text{cavity}) = 1$
Note: **the less specific belief remains valid** after more evidence arrives, but is **not always useful**.
- New evidence may be **irrelevant**, allowing **simplification**:
 $P(\text{cavity}|\text{toothache}, \text{die roll} = 3) = P(\text{cavity}|\text{toothache}) = 0.8$

Conditional probability

- Definition of conditional probability:

$$P(a|b) = \frac{P(a \wedge b)}{P(b)} \text{ if } P(b) \neq 0.$$

- **Product rule** gives an alternative formulation:

$$P(a \wedge b) = P(a|b)P(b) = P(b|a)P(a)$$

- **A general version holds for whole distributions**, e.g.

$$P(\text{Weather}, \text{Cavity}) = P(\text{Weather}|\text{Cavity})P(\text{Cavity})$$

- **Chain rule** is derived by successive application of product rule:

$$\begin{aligned} P(X_1, \dots, X_n) &= P(X_1, \dots, X_{n-1}) P(X_n|X_1, \dots, X_{n-1}) \\ &= P(X_1, \dots, X_{n-2}) P(X_{n-1}|X_1, \dots, X_{n-2}) P(X_n|X_1, \dots, X_{n-1}) \\ &= \dots \\ &= \prod_{i=1}^n P(X_i|X_1, \dots, X_{i-1}) \end{aligned}$$

Bayes Rule

Bayes Rule

$$P(B|A) = \frac{P(A|B)P(B)}{P(A)}$$

Proof.

$$P(A|B)P(B) = P(A, B) = P(B|A)P(A)$$



Thomas Bayes (1701
- 1761)



Bayes Rule (cont'd)

- Useful for assessing **diagnostic** probability from **causal** probability:

$$P(\text{Cause}|\text{Effect}) = \frac{\overbrace{P(\text{Effect}|\text{Cause})}^{\text{Prob(symptoms)}} \overbrace{P(\text{Cause})}^{\text{Prevalence}}}{P(\text{Effect})}$$

- E.g., let **M be meningitis** (acute inflammation of the protective membranes covering the brain / spinal cord), **S be stiff neck**. Assume the doctor knows that the **prevalence of meningitis** is $P(m) = 1/50000$, that the **prior probability of a stiff neck** is $p(s) = 0.01$, and that the **symptom stiff neck** occurs with a probability of **0.7**.

$$P(m|s) = \frac{P(s|m)P(m)}{P(s)} = \frac{0.7 \times 1/50000}{0.01} = 0.0014.$$

- Note: the posterior probability of meningitis is still very small (1 in 700 patients)!

Bayes rule (cont'd)

Question: Why should it be easier to estimate the conditional probabilities in the causal direction $P(\text{Effect}|\text{Cause})$, as compared to the diagnostic direction, $P(\text{Cause}|\text{Effect})$?

There are two possible answers (in a medical setting):

- We might have access to a **collection of health records** for patients having meningitis $\rightsquigarrow$ **we can estimate $P(s|m)$.**

For directly estimating $P(m|s)$ we would need a database of all cases of the **very unspecific symptom**.

- Diagnostic knowledge might be **more fragile** than causal knowledge.
 - ▶ Assume a doctor has directly estimated $P(m|s)$.
Sudden epidemic $\rightsquigarrow P(m)$ will go up...but how to update $P(m|s)$?
 - ▶ Other doctor uses Bayes rule, he knows that $P(m|s) \propto p(s|m)p(m)$ should go up proportionately with $p(m)$.
Note: **Causal information $P(s|m)$ is unaffected by the epidemic** (it simply reflects the way how meningitis works)!

Origins of probabilities

Historically speaking, probabilities have been regarded in a number of different ways:

- **Frequentist position:** probabilities come from **measurements**.
 - ▶ The assertion $P(\text{cavity}) = 0.05$ means that 0.05 is the fraction that would be observed in the limit of infinitely many samples.
 - ▶ From a finite sample, we can estimate this true fraction and also the accuracy of this estimate.
- **Objectivist view:** probabilities are actual **properties of the universe**
 - ▶ An excellent example: quantum phenomena.
 - ▶ A less clear example: coin flipping – the uncertainty is probably due to our uncertainty about the initial conditions of the coin.

Origins of probabilities

- **Subjectivist view**: probabilities are an **agent's degrees of belief**, rather than having any external physical significance.
- The **Bayesian view** allows any self-consistent ascription of prior probabilities to propositions, but then insists on proper **Bayesian updating** as evidence arrives.

For example $P(\text{cavity}) = 0.05$ denotes the degree of belief that a random person has a cavity **before we make any actual observation** of that person.

Updating in the light of **further evidence** “person has a *toothache*”:

$$P(\text{cavity}|\text{toothache}) = \alpha P(\text{toothache}|\text{cavity})P(\text{cavity})$$

The reference class problem

- Bayesian viewpoint is often criticized because of the use of subjective believes...
...**but even a strict frequentist position involves subjective analysis!**
- **Example:** Say a doctor takes a frequentist approach to diagnosis. She examines a large number of people to establish the probability of whether or not they have heart disease.
- To be accurate she tries to measure “similar people” (she knows for example that gender might be important) $\rightsquigarrow$ “reference class”.
- ...but probably other variables might also be important...
- Taken to an extreme, all people are different
 $\rightsquigarrow$ **reference class is empty.**
- Some subjective assumptions must be involved in the design of nonempty reference classes...a tricky problem in the philosophy of science.

Common continuous distributions: Normal

- The **pdf** of the **normal distribution** is

$$p(x \mid \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}},$$

where μ is the **mean**, σ^2 is the **variance**.

The **inverse variance** is sometimes called **precision**.

- The **cdf** of the **standard normal distribution** is the integral

$$\Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-t^2/2} dt.$$

It has no closed form expression.

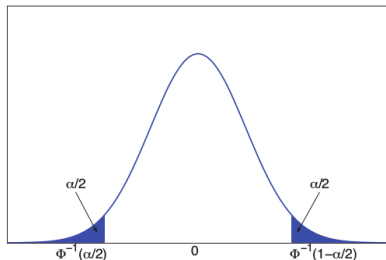
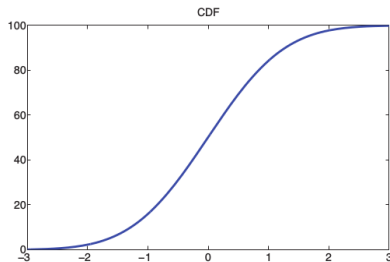
- The **cdf** is sometimes expressed in terms of the **error function**

$$\text{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt,$$

as follows:

$$\Phi(x) = \frac{1}{2} \left[1 + \text{erf} \left(\frac{x}{\sqrt{2}} \right) \right].$$

Probability for continuous variables



Left: **cdf** for the standard normal, $\mathcal{N}(0, 1)$. Right: corresponding **pdf**.

- Shaded regions each contain $\alpha/2$ of the probability mass
 $\rightsquigarrow$ nonshaded region contains $1 - \alpha$.
- Left cutoff point is $\Phi^{-1}(\alpha/2)$, Φ is cdf of standard Gaussian.
- By symmetry, the right cutoff point is $\Phi^{-1}(1 - \alpha/2) = -\Phi^{-1}(\alpha/2)$.
- If $\alpha = 0.05$, the central interval is 95%, left cutoff is -1.96 , right cutoff is 1.96 .

Common continuous distributions: Normal

- If σ tends to zero, $p(x)$ tends to zero at any $x \neq \mu$, but grows without limit if $x = \mu$, while its integral remains equal to 1.
- Can be defined as a generalized function: **Dirac's delta function** δ translated by the mean: $p(x) = \delta(x - \mu)$, where

$$\delta(x) = \begin{cases} +\infty, & x = 0 \\ 0, & x \neq 0, \end{cases}$$

additionally constrained to satisfy the identity

$$\int_{-\infty}^{\infty} \delta(x) dx = 1.$$

- **Sifting property:** selecting out a single term from an integral:

$$\int_{-\infty}^{\infty} f(x) \delta(x - z) dx = f(z)$$

since the integrand is only non-zero if $x - z = 0$.

Central Limit Theorem

- Sums of independent normal RVs always have a normal distribution.
- Under certain (fairly common) conditions, the **sum of many i.i.d. RVs** will have an **approximately normal distribution.**
- Let $X_1, \dots, X_n$ be i.i.d. RVs with the same (arbitrary) distribution, zero mean, and variance σ^2 .

- Let

$$Z = \sqrt{n} \left(\frac{1}{n} \sum_{i=1}^n X_i \right)$$

- Then, as n increases, the probability distribution of Z will tend to the **normal distribution** with zero mean and variance σ^2 .

Frequentist and Bayesian view: a simple example

- Assume $x_1, \dots, x_n$ are drawn i.i.d. from normal $\mathcal{N}(\mu, \sigma^2)$ with known variance σ^2 . What can be said about μ ?
- Frequentist view: no further probabilistic assumptions
 $\leadsto$ **treat μ as an unknown constant.**

- Theorem:** Let X and Y be independent normal RVs, then their sum is also normally distributed. i.e., if

$$X \sim \mathcal{N}(\mu_X, \sigma_X^2), \quad Y \sim \mathcal{N}(\mu_Y, \sigma_Y^2)$$

$$Z = X + Y, \text{ then } Z \sim \mathcal{N}(\mu_X + \mu_Y, \sigma_X^2 + \sigma_Y^2).$$

- Theorem:** If $y = c + bx$ is an **affine transformation** of

$$X \sim \mathcal{N}(\mu, \sigma^2), \text{ then } Y \sim \mathcal{N}(c + b\mu, b^2\sigma^2)$$

- The **sample mean** $\bar{x} = \sum_i x_i / n$ is the observed value of the RV
 $\bar{X} \sim \mathcal{N}(\mu, \bar{\sigma}^2), \quad \text{with } \bar{\sigma}^2 = n\sigma^2 / n^2 = \sigma^2 / n.$

Frequentist and Bayesian view: a simple example

- Now define the **transformed random variable**

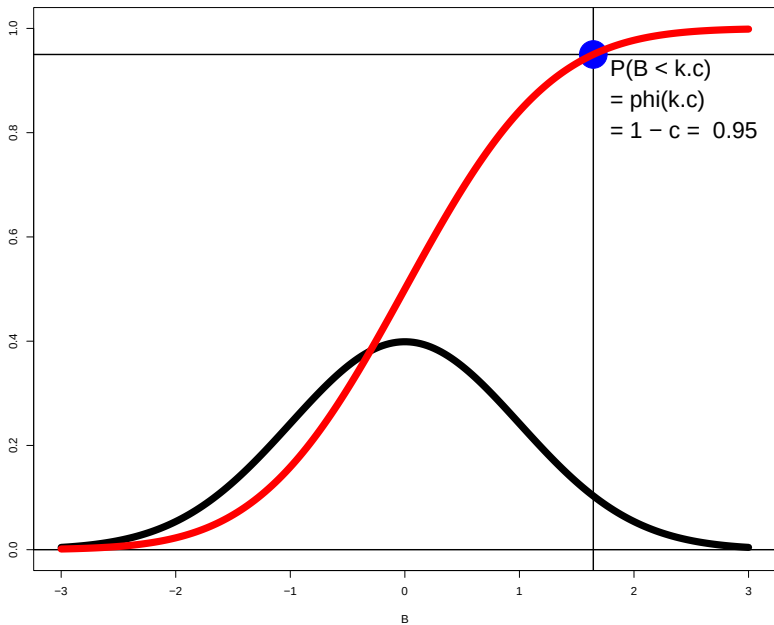
$$B := \frac{\mu - \bar{X}}{\bar{\sigma}} \sim \mathcal{N}(0, 1), \text{ (i.e. standard normal).}$$

- Use normal cdf $\Phi(k_c) = P(B < k_c)$ to derive an **upper limit for μ** :

$$\begin{aligned} P(B < k_c) &= \Phi(k_c) = 1 - c \\ &= P(-\bar{\sigma}B > -\bar{\sigma}k_c) \\ &= P(\underbrace{\mu - \bar{\sigma}B}_{\bar{X}} > \mu - \bar{\sigma}k_c) \\ &= P(\bar{X} + \bar{\sigma}k_c > \mu). \end{aligned}$$

$$\Rightarrow P(\mu < \bar{X} + \bar{\sigma}k_c) = 1 - c$$

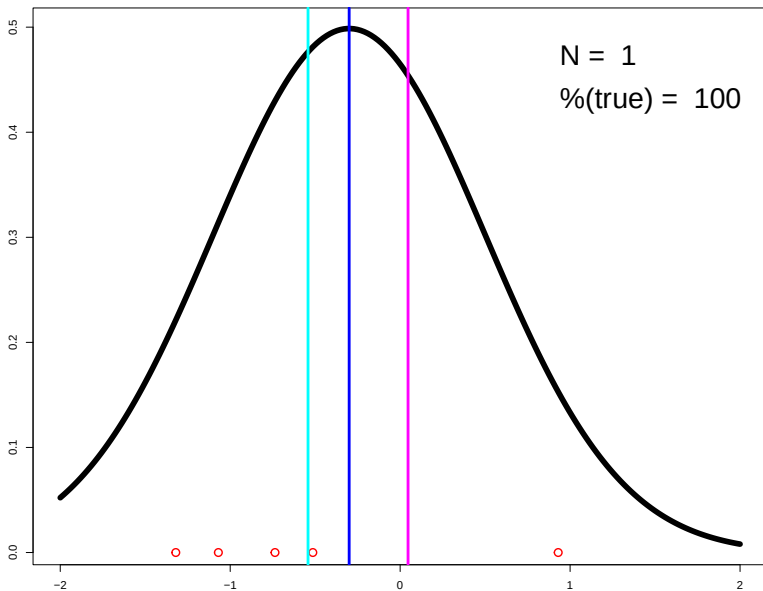
- Define $B' = -B \rightsquigarrow$ **lower limit.**



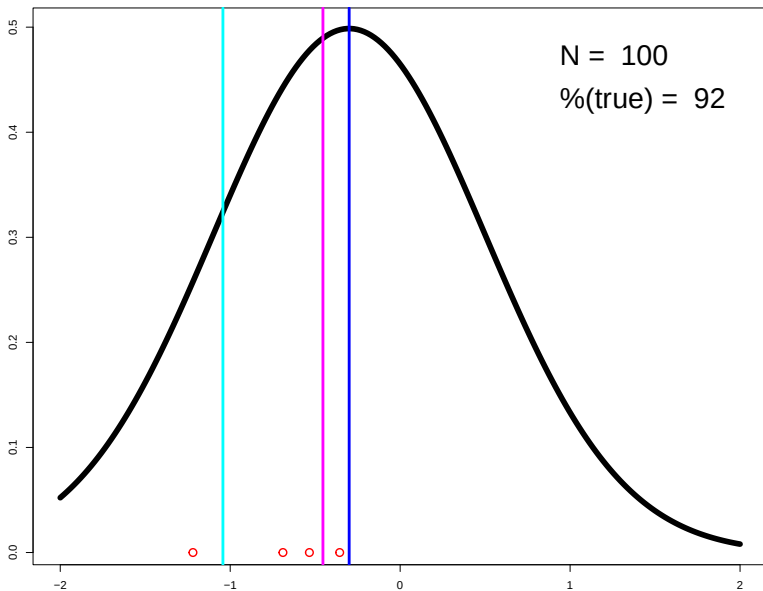
Frequentist and Bayesian view: a simple example

- The statement $P(\mu < \bar{X} + \bar{\sigma}k_c) = 1 - c$ can be interpreted as specifying a **hypothetical long run of statements** about the **constant μ , a portion $1 - c$ of which is correct.**
- Note that $\bar{X}$ is a RV that takes **one specific value $\bar{x}$** for one dataset of n observations $\{x_1, \dots, x_n\}$.
- For the **actually observed $\bar{x}$** , the statement $\mu < \bar{x} + \bar{\sigma}k_c$ can be interpreted as **one** of a long run of such statements about μ .
- Arguments involving probability **only via its (hypothetical) long-run frequency interpretation** are called **frequentist**.
- In the frequentist world we define procedures for assessing evidence that are calibrated by how they would perform **were they used repeatedly**.

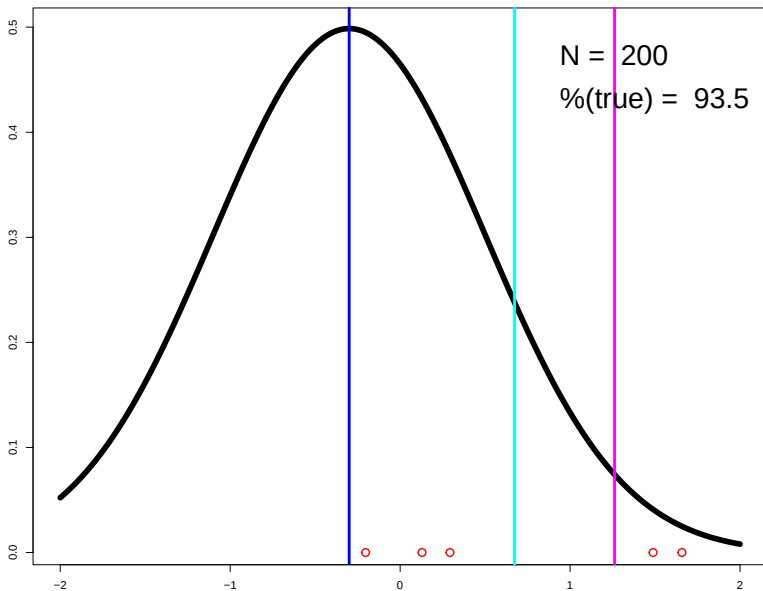
$$P(\mu < \bar{x} + \text{sig.bar} * k.c) = 1 - c = 0.95$$



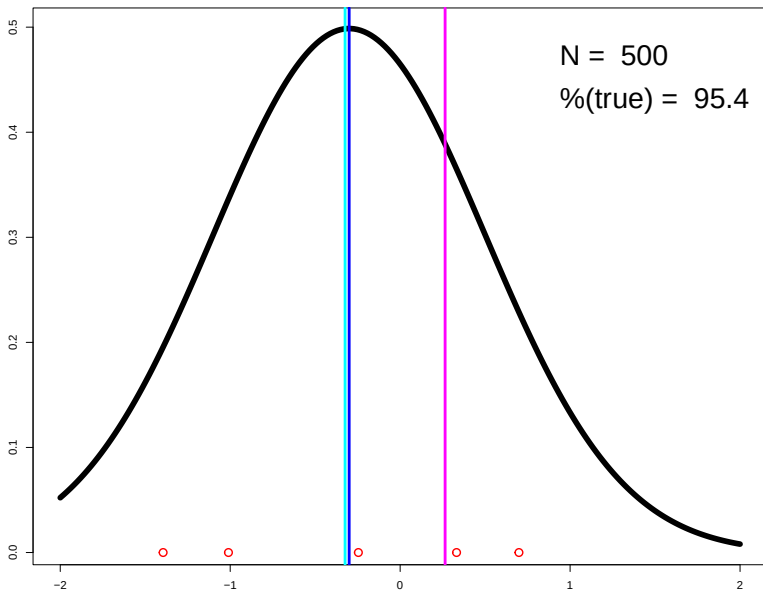
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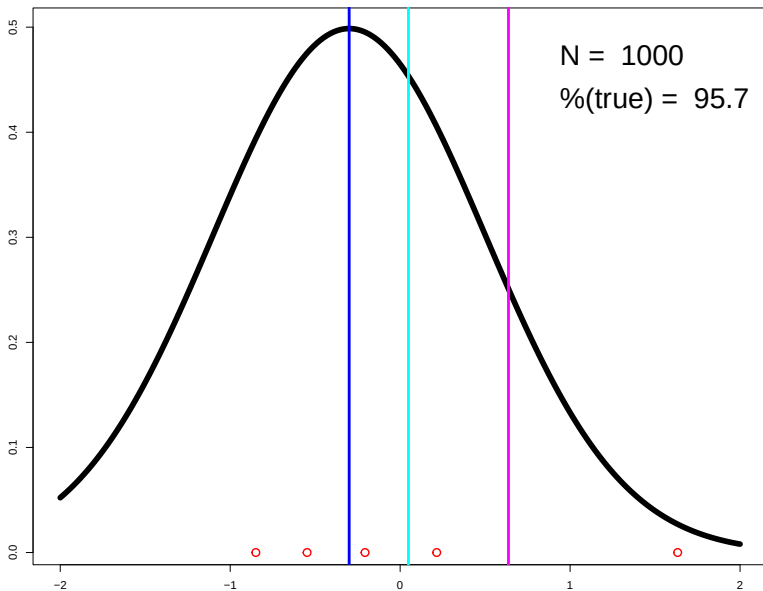
$$P(\mu < \bar{x} + \text{sig.bar} * k.c) = 1 - c = 0.95$$



$$P(\mu < \bar{x} + \bar{s} \cdot k.c) = 1 - c = 0.95$$



$$P(\mu < \bar{x} + \text{sig.bar} * k.c) = 1 - c = 0.95$$



Frequentist and Bayesian view: a simple example

- From the Bayesian viewpoint, we treat μ as having a probability distribution **both with and without the data**:

$\rightsquigarrow$ **treat μ as a random variable.**

- Bayes' theorem: $p(\mu|\bar{x}) \propto p(\bar{x}|\mu)p(\mu)$.
- Intuitive idea:
 - all relevant information about μ is in the conditional distribution, given the data;
 - this distribution is determined by the elementary formulae of probability theory;
 - remaining problems are solely computational.
- Example: choose $p(\mu) = \mathcal{N}(m, \nu^2) \rightsquigarrow p(\mu|x) = \mathcal{N}(\tilde{m}, \tilde{\nu}^2)$ with

$$\tilde{m} = \frac{\bar{x}/\bar{\sigma}^2 + m/\nu^2}{1/\bar{\sigma}^2 + 1/\nu^2}, \quad \tilde{\nu}^2 = \frac{1}{1/\bar{\sigma}^2 + 1/\nu^2}$$

“Normal likelihood times normal prior gives normal posterior”

Frequentist and Bayesian view: a simple example

- Same reasoning as before: define transformed $\tilde{B} := \frac{\mu - \tilde{m}}{\tilde{\nu}} \sim \mathcal{N}(0, 1)$
- Upper limit for μ : $P(\mu < \tilde{m} + k_c \tilde{\nu}) = 1 - c$.
- If the **prior variance** $\nu^2 \gg \bar{\sigma}^2$ and the **prior mean** m is not too different from $\bar{X}$, this limit agrees closely with the one obtained by the frequentist method (because then $\tilde{m} \approx \bar{x}$ and $\tilde{\nu} \approx \bar{\sigma}$).
- Note that this approximation becomes exact in the limit as $n \rightarrow \infty$, since then $\bar{\sigma}^2 = \sigma^2/n \rightarrow 0$.
- This broad parallel is in no way specific to the normal distribution (mainly due to the **central limit theorem**).
- **Warning:** there **are** situations in which there are fundamental differences!
- See the discussion of the “likelihood principle” in https://en.wikipedia.org/wiki/Likelihood_principle, or the paper “The Interplay of Bayesian and Frequentist Analysis” by M. J. Bayarri and J. O. Berger, or the book (D.R. Cox, Principles of statistical inference, Cambridge, 2006).